

ECE 105: Introduction to Electrical Engineering

Lecture 16

Sensors 3

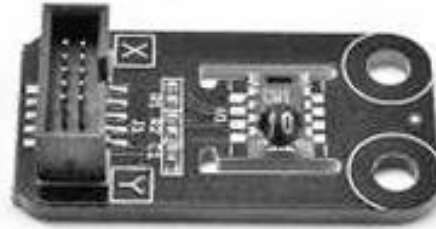
Yasser Khan

Rehan Kapadia

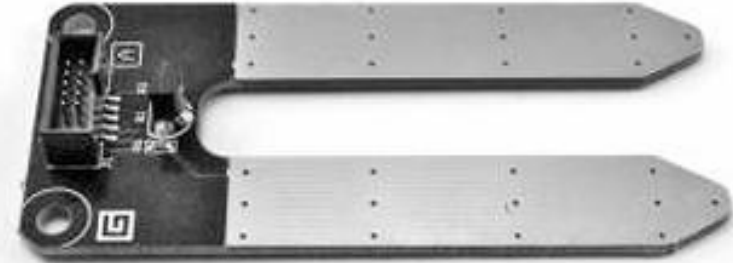
Other sensors



Gas Sensor

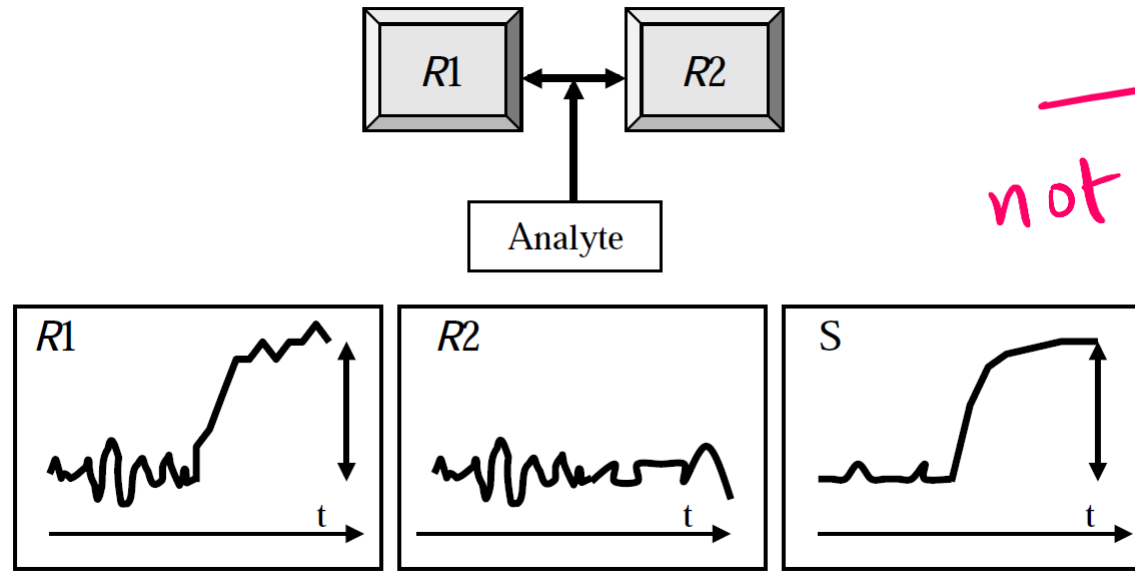


Humidity



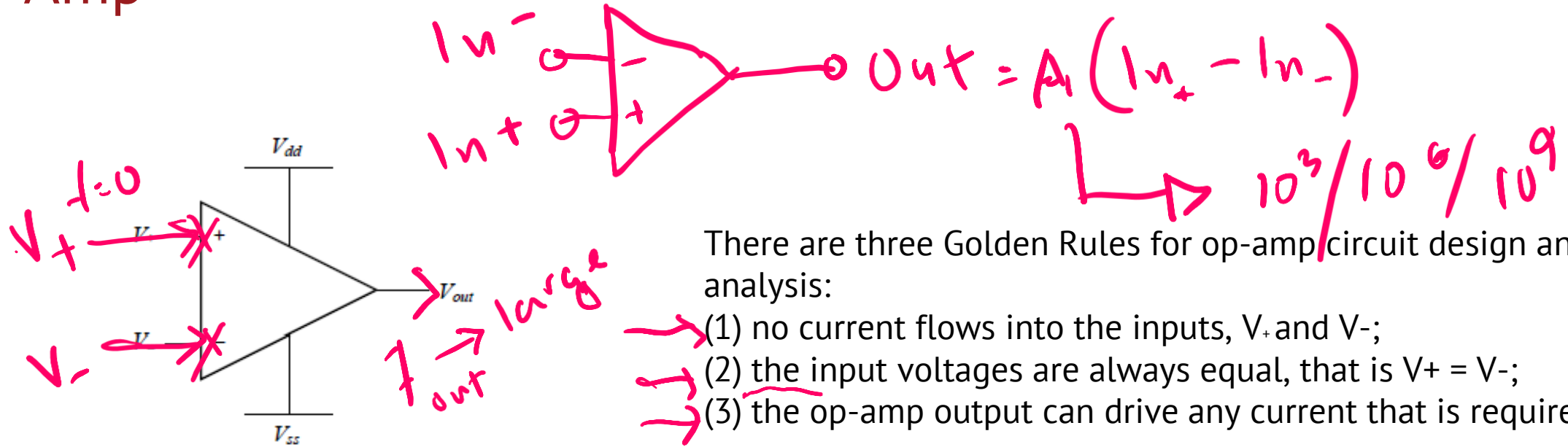
Moisture

Differential measurement



sensing
→
not sensing
→
(S-NS)

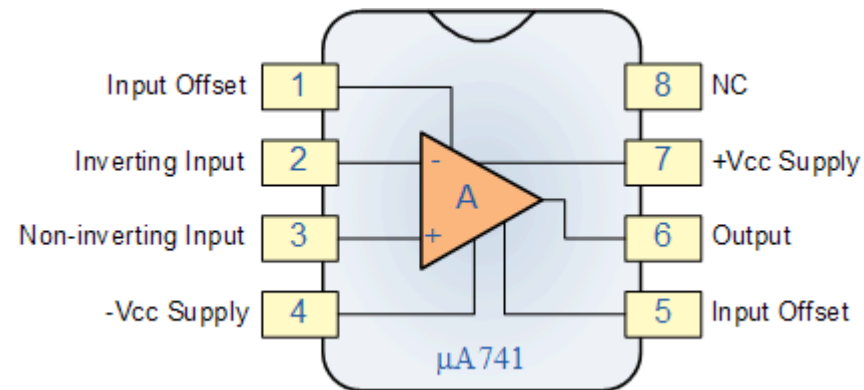
Op Amp



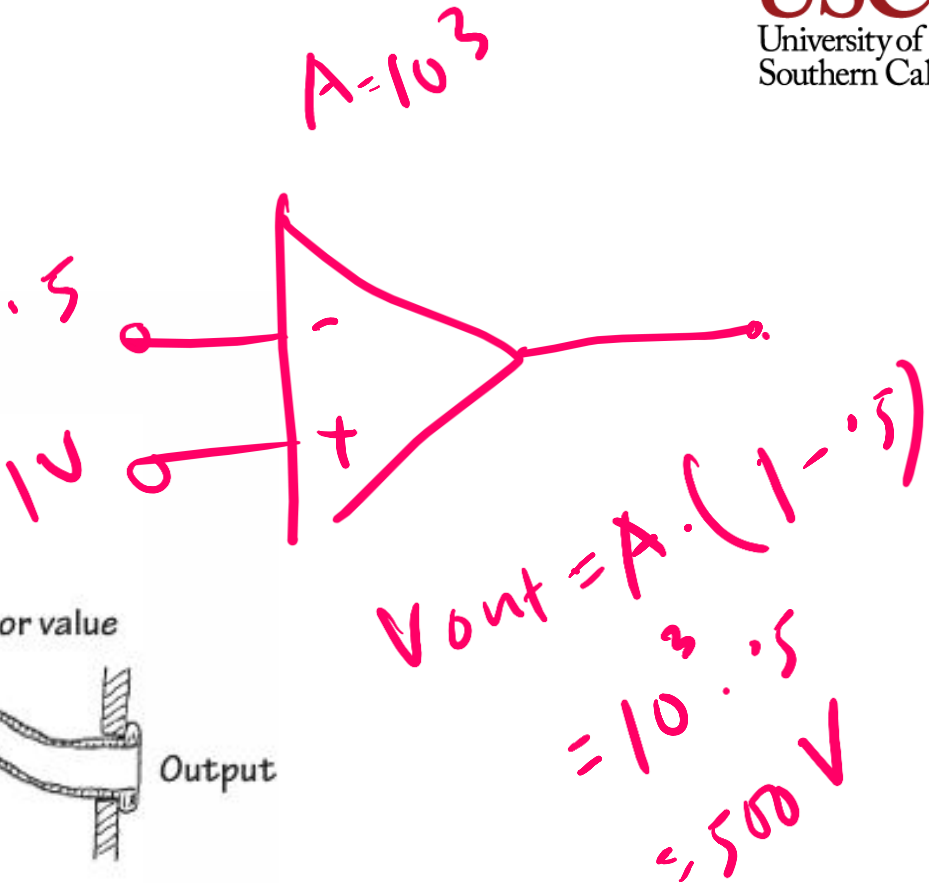
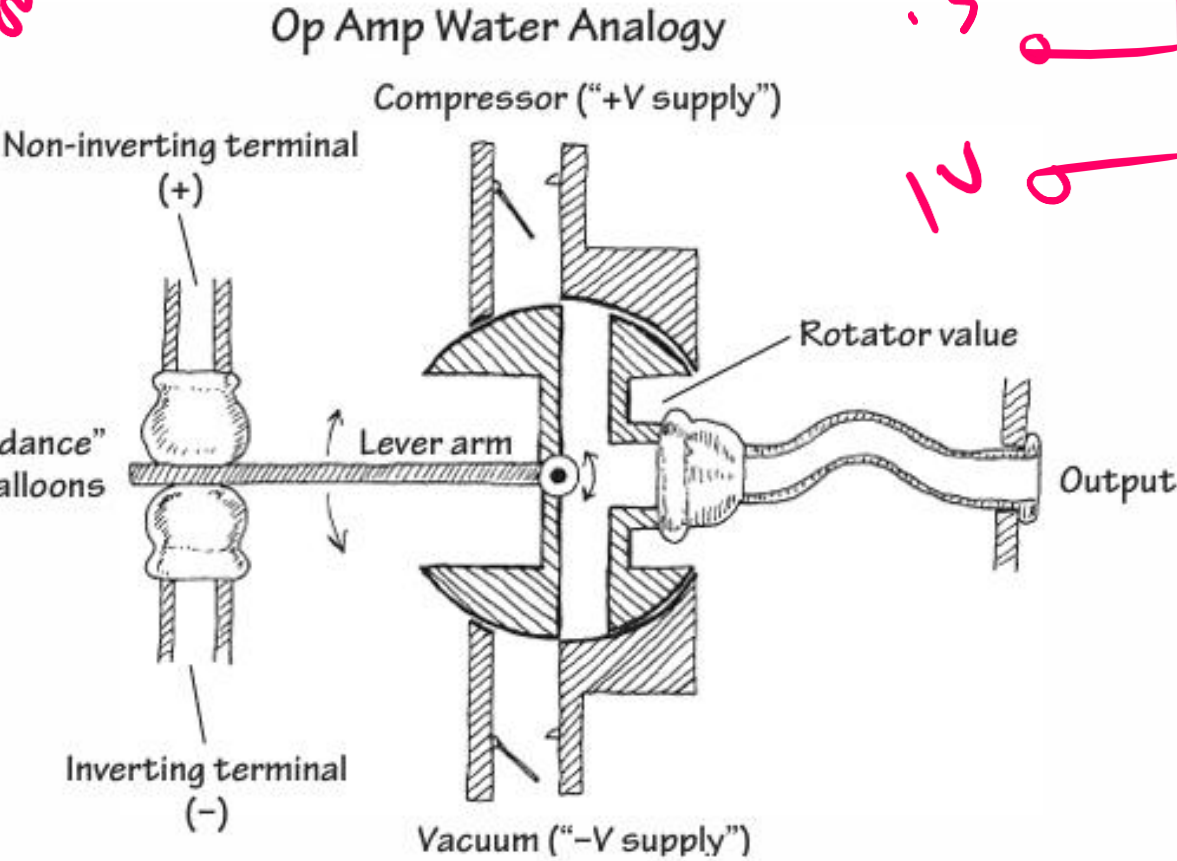
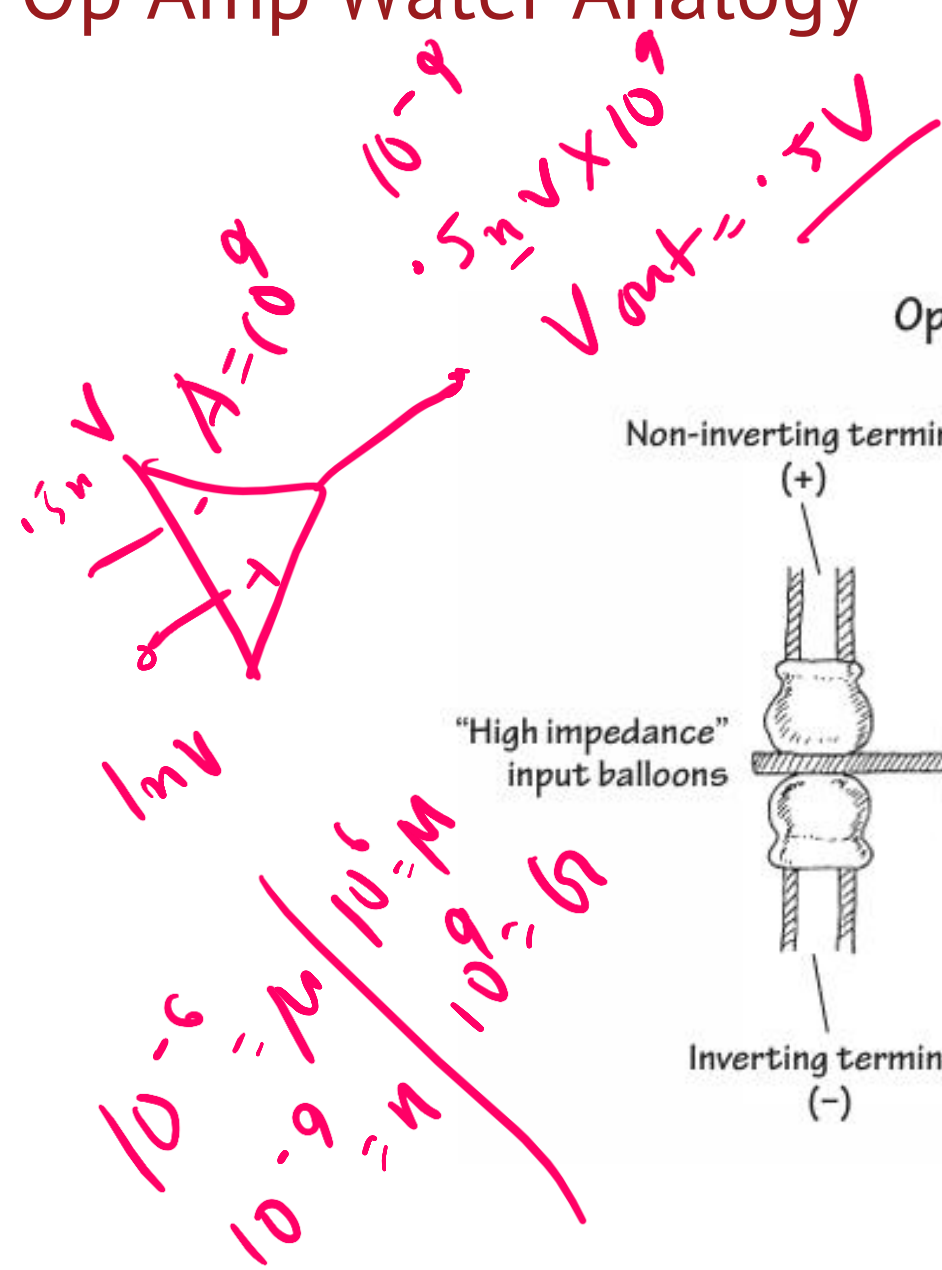
There are three Golden Rules for op-amp/circuit design and analysis:

- (1) no current flows into the inputs, V_+ and V_- ;
- (2) the input voltages are always equal, that is $V_+ = V_-$;
- (3) the op-amp output can drive any current that is required.

Circuit schematic symbol for an operational amplifier.

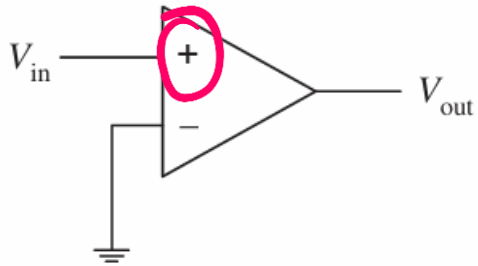


Op Amp Water Analogy

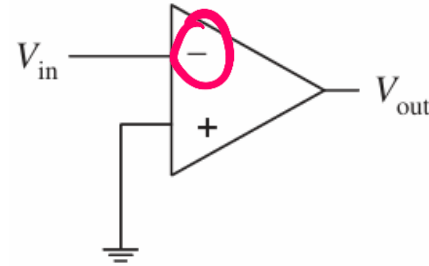


Non-inverting and inverting

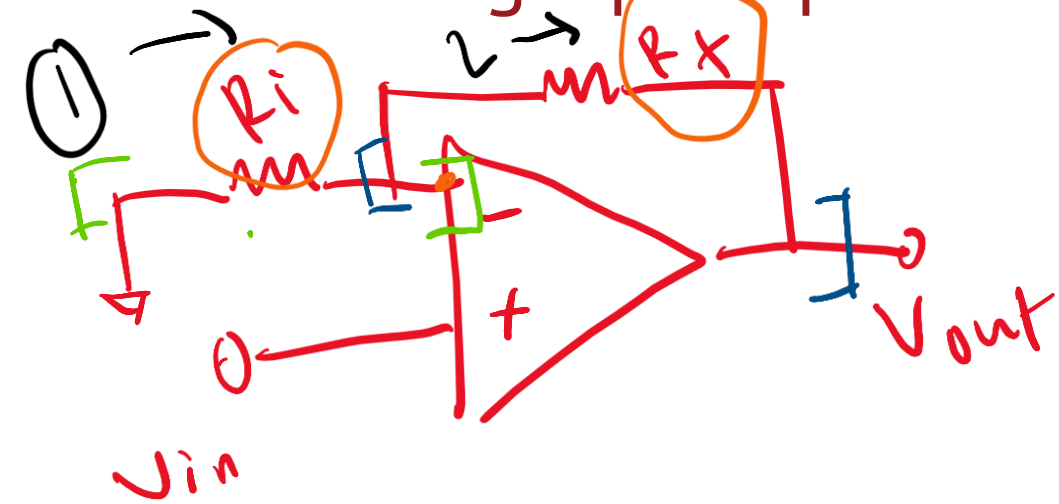
Non-inverting



Inverting



Non-inverting Op Amp

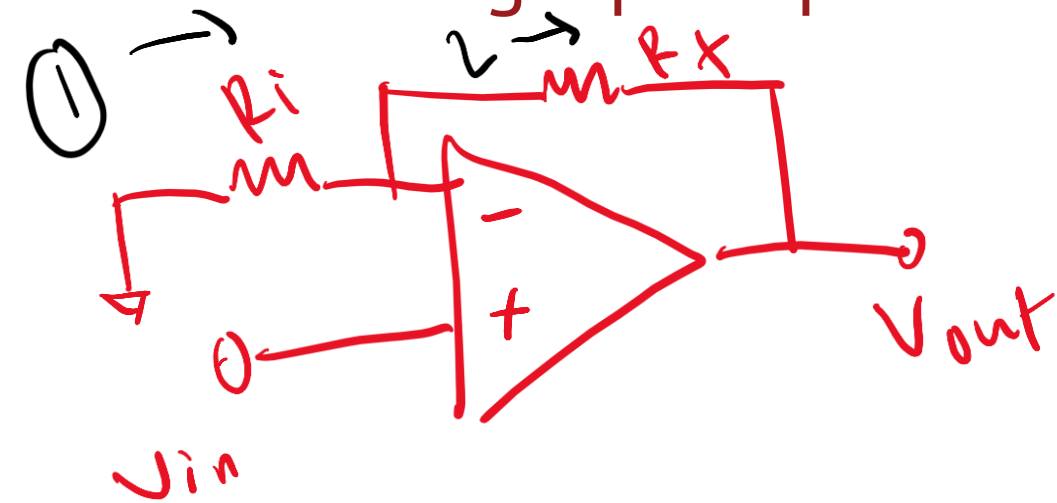


$R_i = R$ input
 $R_f = R$ feedback

Green

$$\frac{0 - V_{in}}{R_i} = \frac{\text{Blue } V_{in} - V_{out}}{R_f}$$

Non-inverting Op Amp



②

$$\frac{0 - V_{in}}{R_i} = \frac{V_{in} - V_{out}}{R_f}$$

$$\frac{-V_{in}}{R_i} = \frac{V_{in}}{R_f} - \frac{V_{out}}{R_f}$$

$$\frac{-V_{out}}{R_f} = \frac{-V_{in}}{R_i} - \frac{V_{in}}{R_f}$$

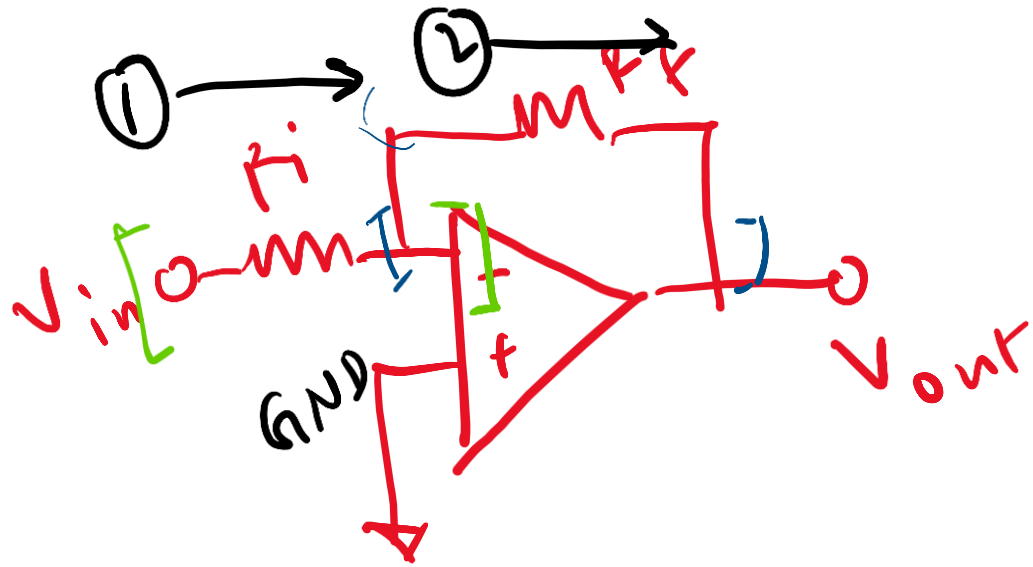
$$V_{out} = V_{in} \cdot R_f \cdot \left(\frac{R_i + R_f}{R_i \cdot R_f} \right)$$

$$V_{out} = V_{in} \left(\frac{R_i + R_f}{R_i} \right)$$

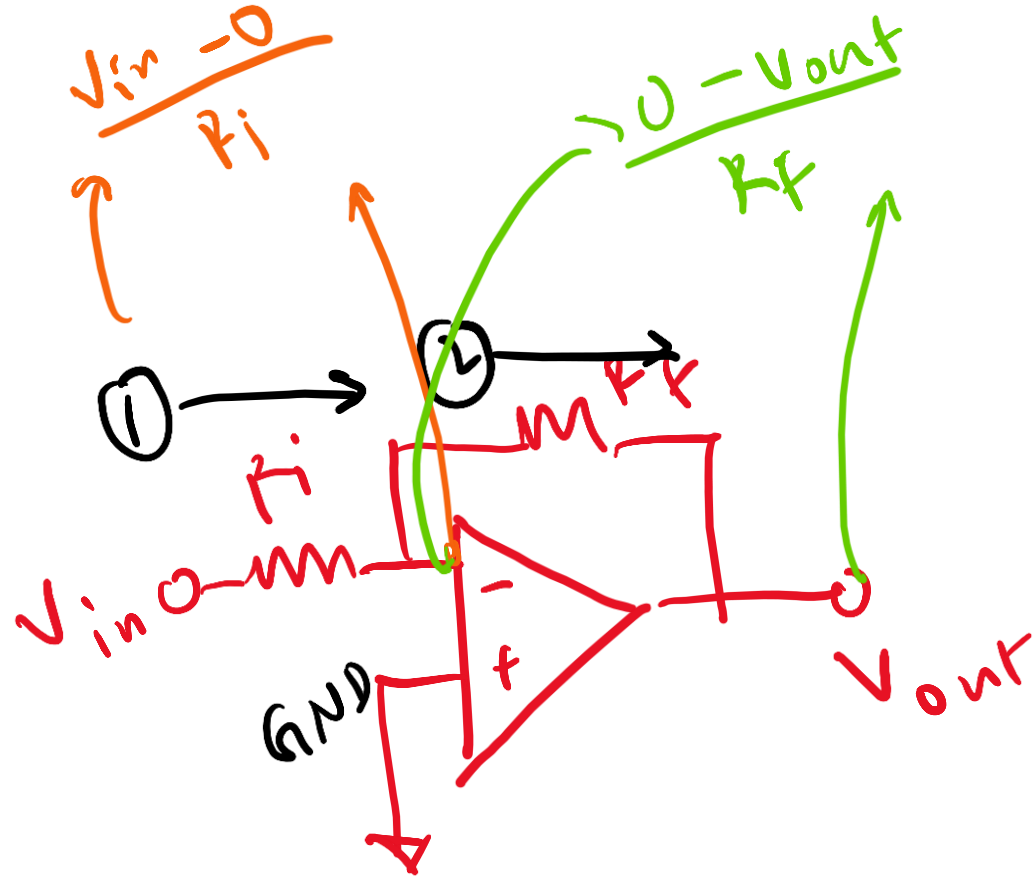
$$V_{out} = A \cdot V_{in}$$

Inverting Op Amp

$$\frac{V_{in} - 0}{R_i} = \frac{0 - V_{out}}{R_f}$$



Inverting Op Amp



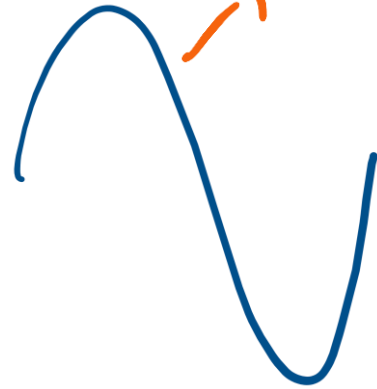
$$\textcircled{1} \frac{V_{in} - 0}{R_i} = \textcircled{2} \frac{0 - V_{out}}{R_f}$$

$$\frac{V_{in}}{R_i} = \frac{-V_{out}}{R_f}$$

$$V_{out} = \left(\frac{-R_f}{R_i} \right) V_{in}$$

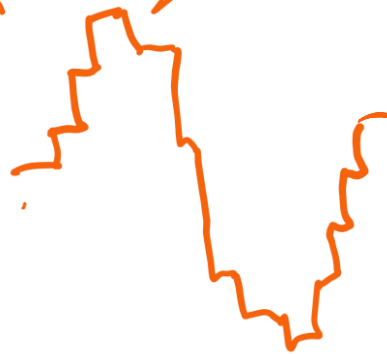
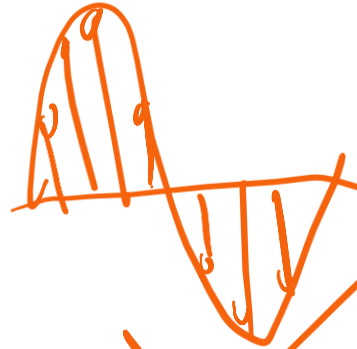
$$V_{out} = (-A) V_{in}$$

Analog



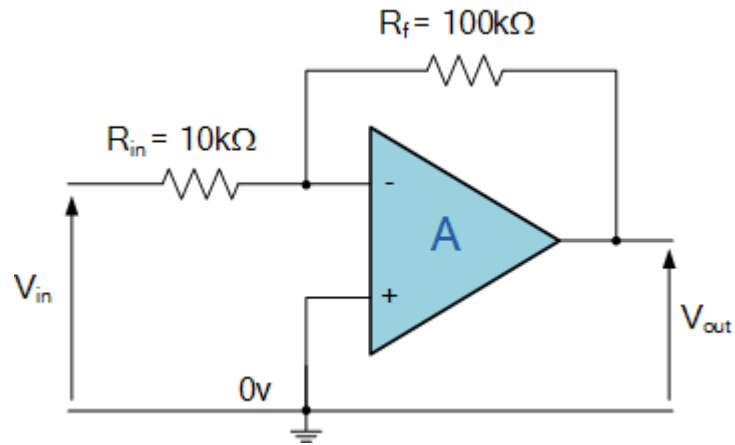
4.9372111...

Digital



1010010000

Example



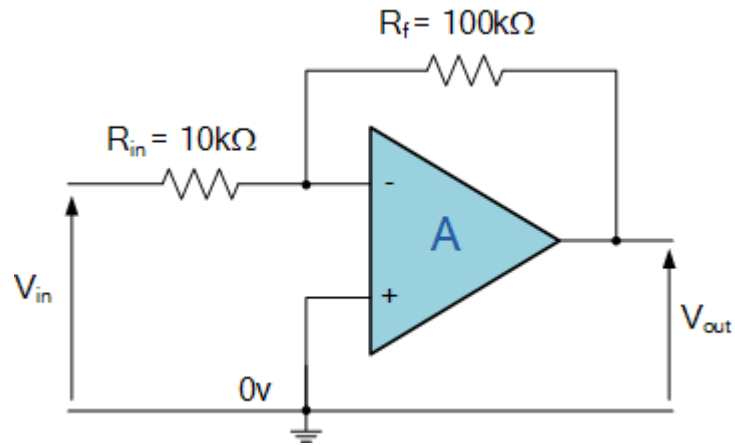
inverting

$$V_{out} = \left(\frac{-R_f}{R_i} \right) V_{in}$$

$$= \left(-\frac{100K}{10K} \right) V_{in}$$

$$= (-10) V_{in}$$

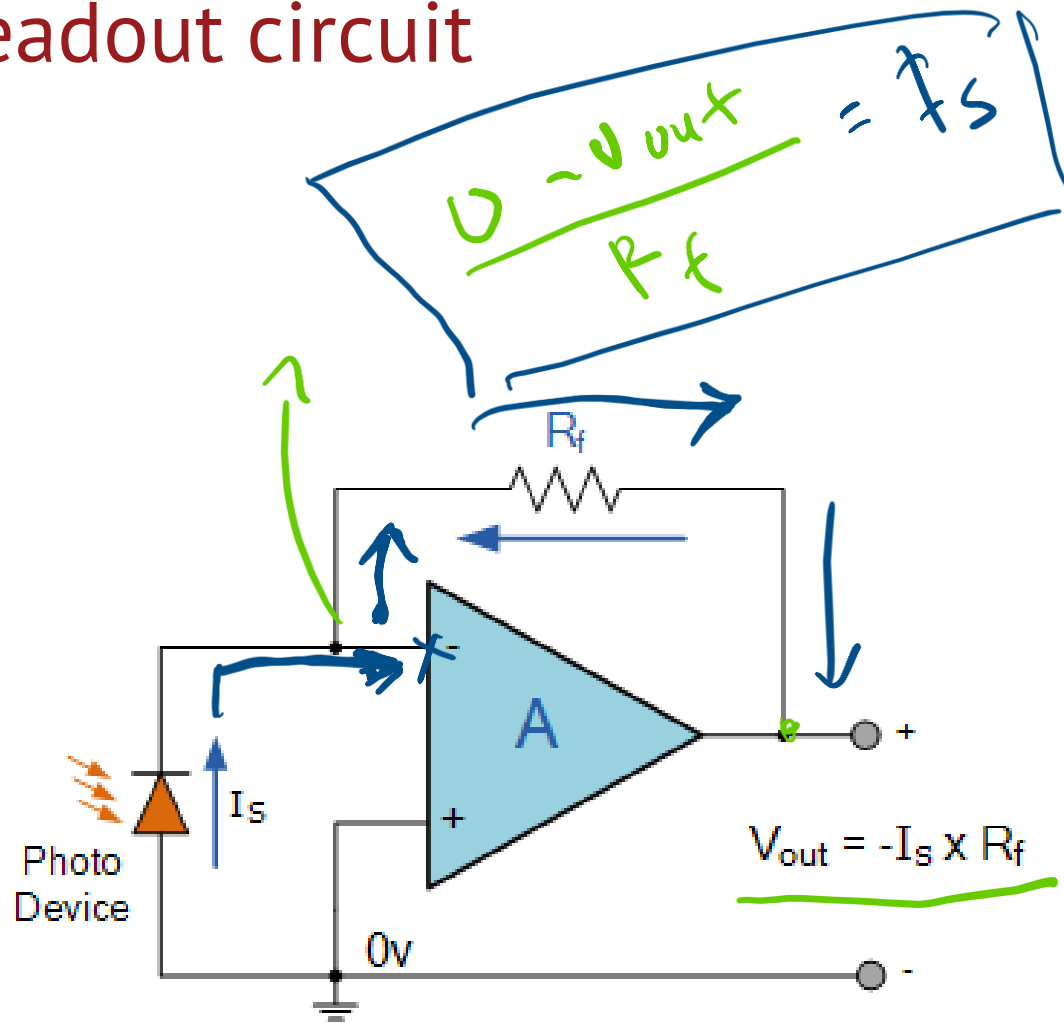
Example



$$V_{out} = -\frac{R_f}{R_i} \cdot V_{in}$$
$$= -\frac{100k}{10k} V_{in}$$

$$V_{out} = (-10) V_{in}$$

PD readout circuit



$$\frac{0 - V_{out}}{R_f} = I_s$$

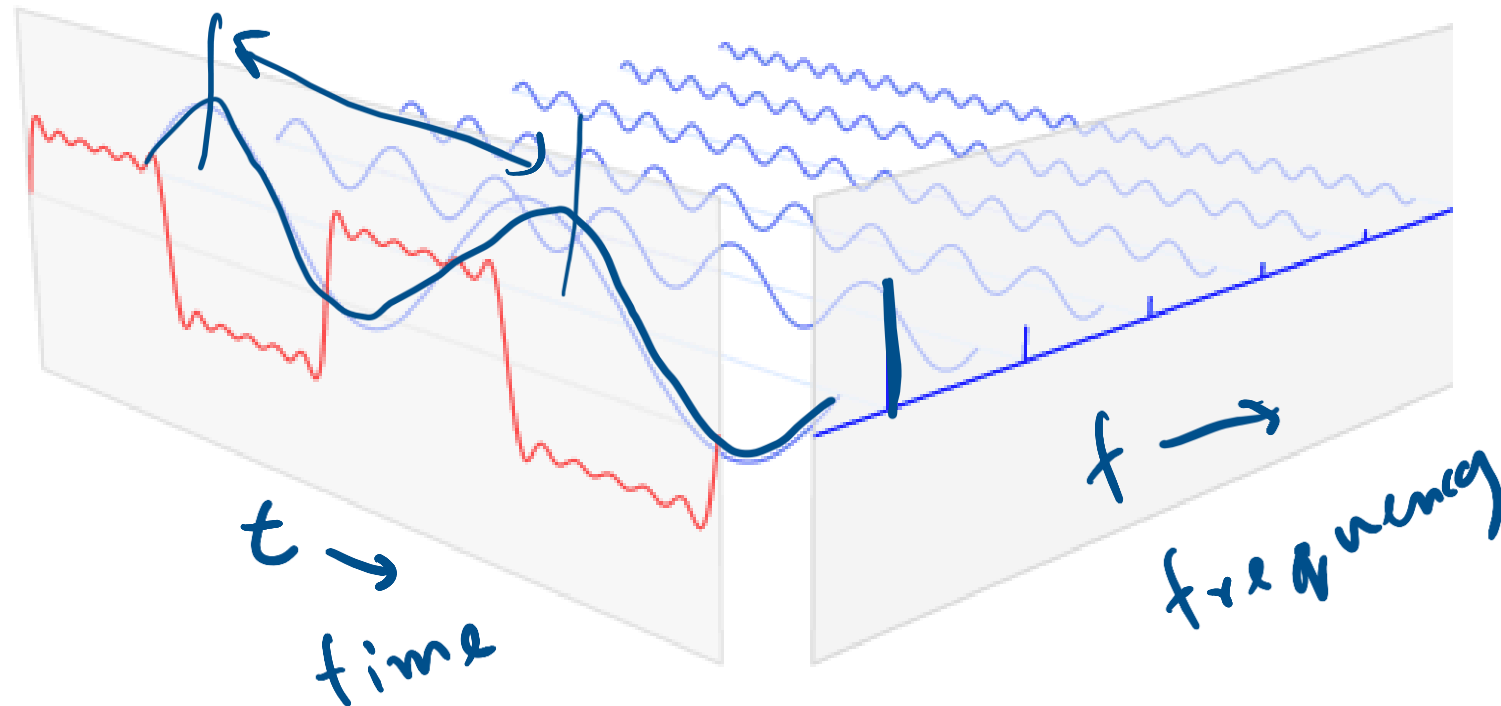
$$V_{out} = -I_s \cdot R_f$$

We need V_{out} ($-3V$). $I_s = 10\mu A$ $R_f = \frac{3}{10\mu} = \frac{3}{10} \times 10^6 \Omega = 300k\Omega$

What R_f we should use?

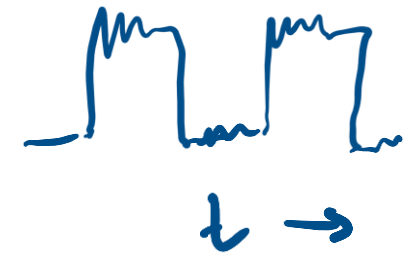
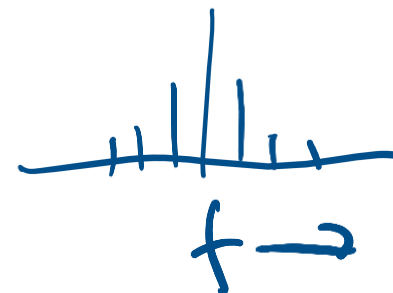
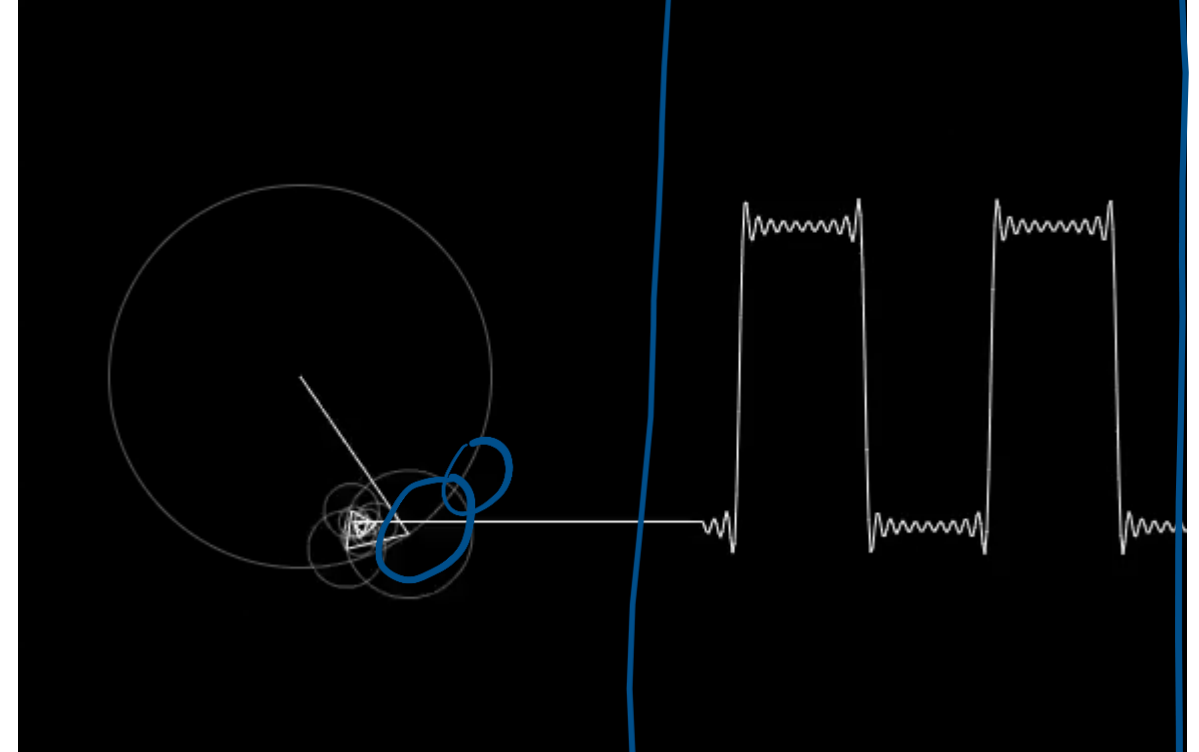
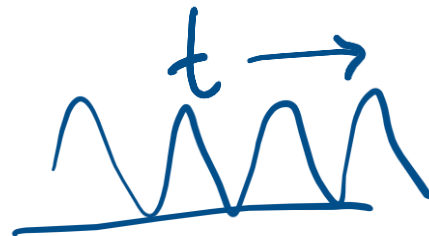
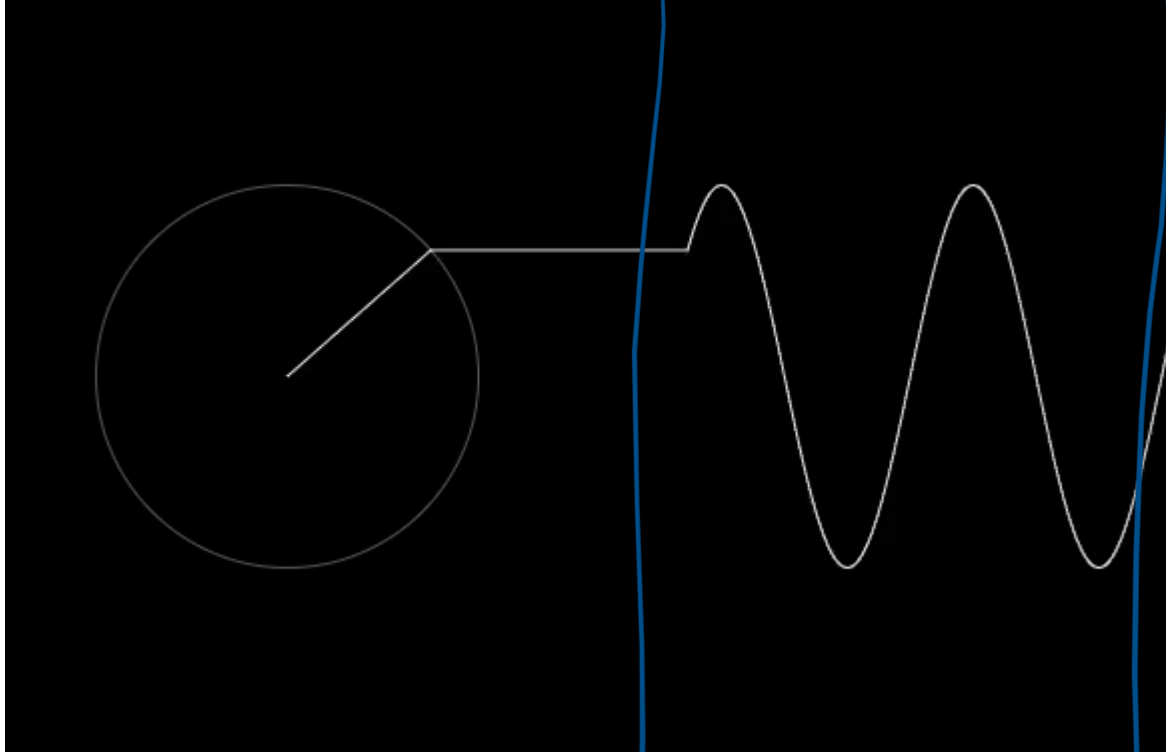
Time domain to frequency domain

- The Fourier transform is an analysis process, decomposing a complex-valued function into its constituent frequencies and their amplitudes.

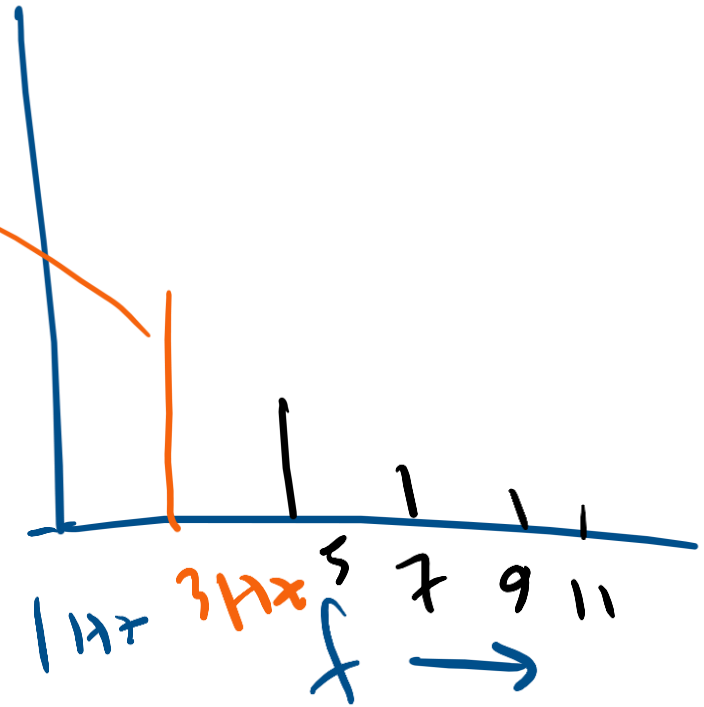
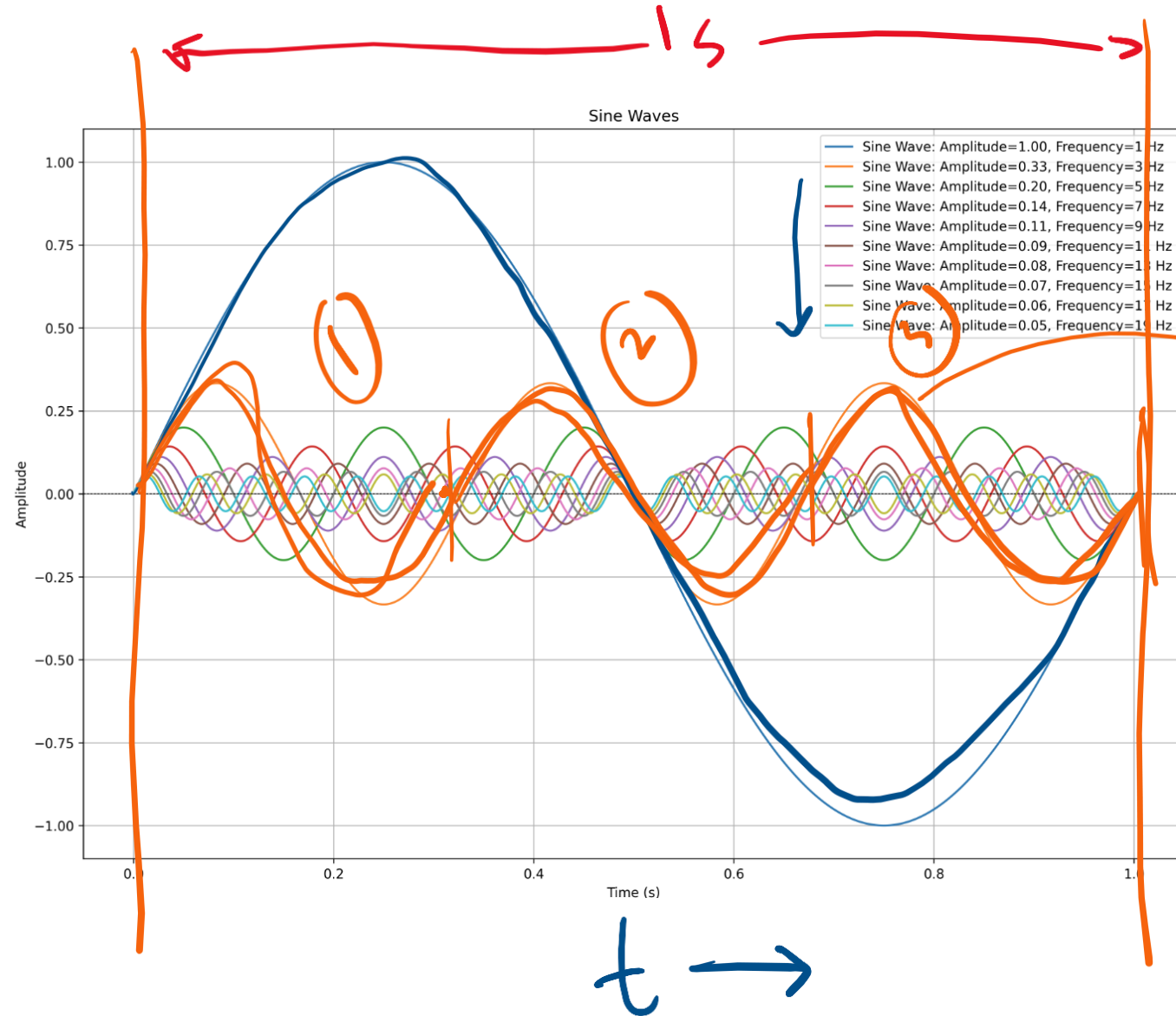


Time domain to frequency domain

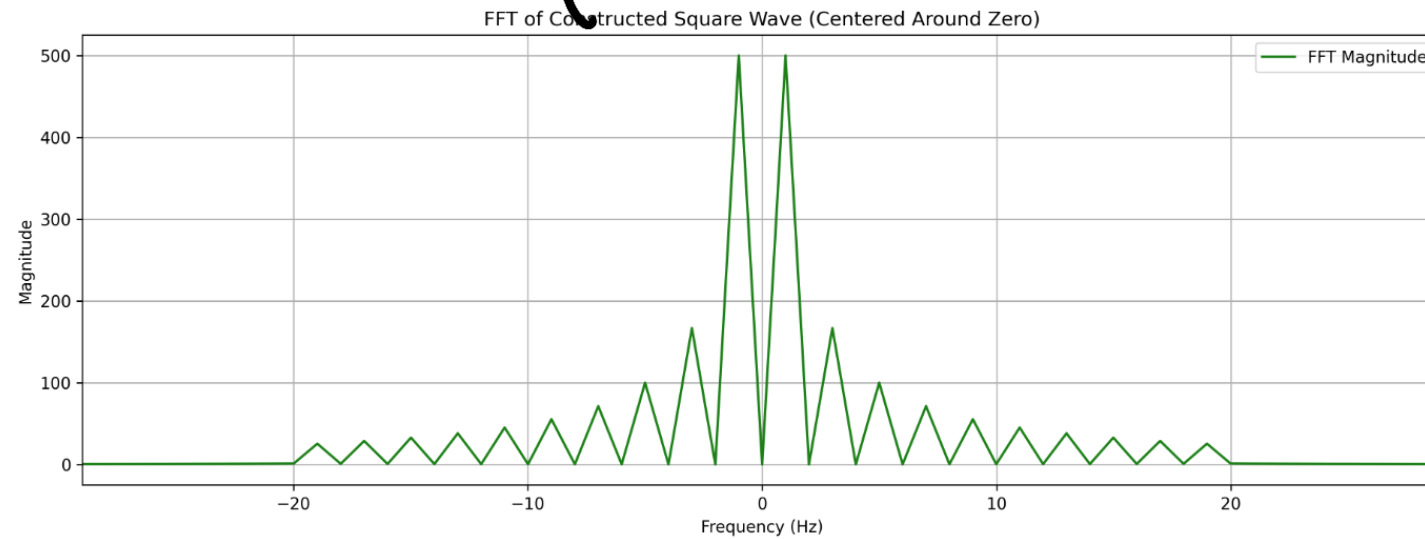
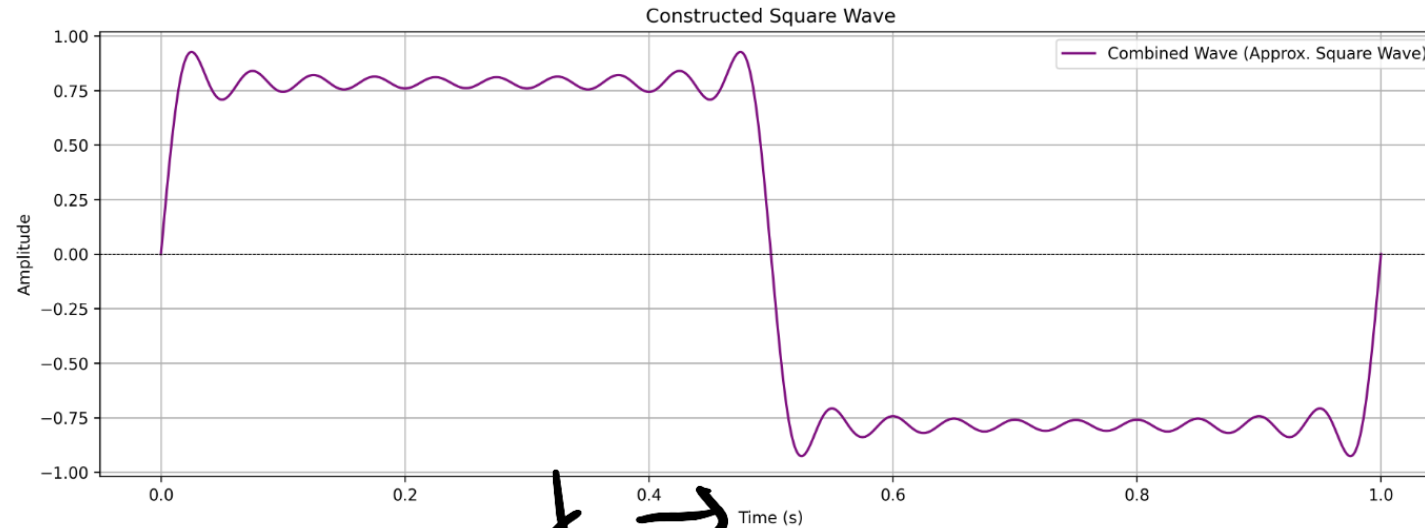
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Time domain to frequency domain

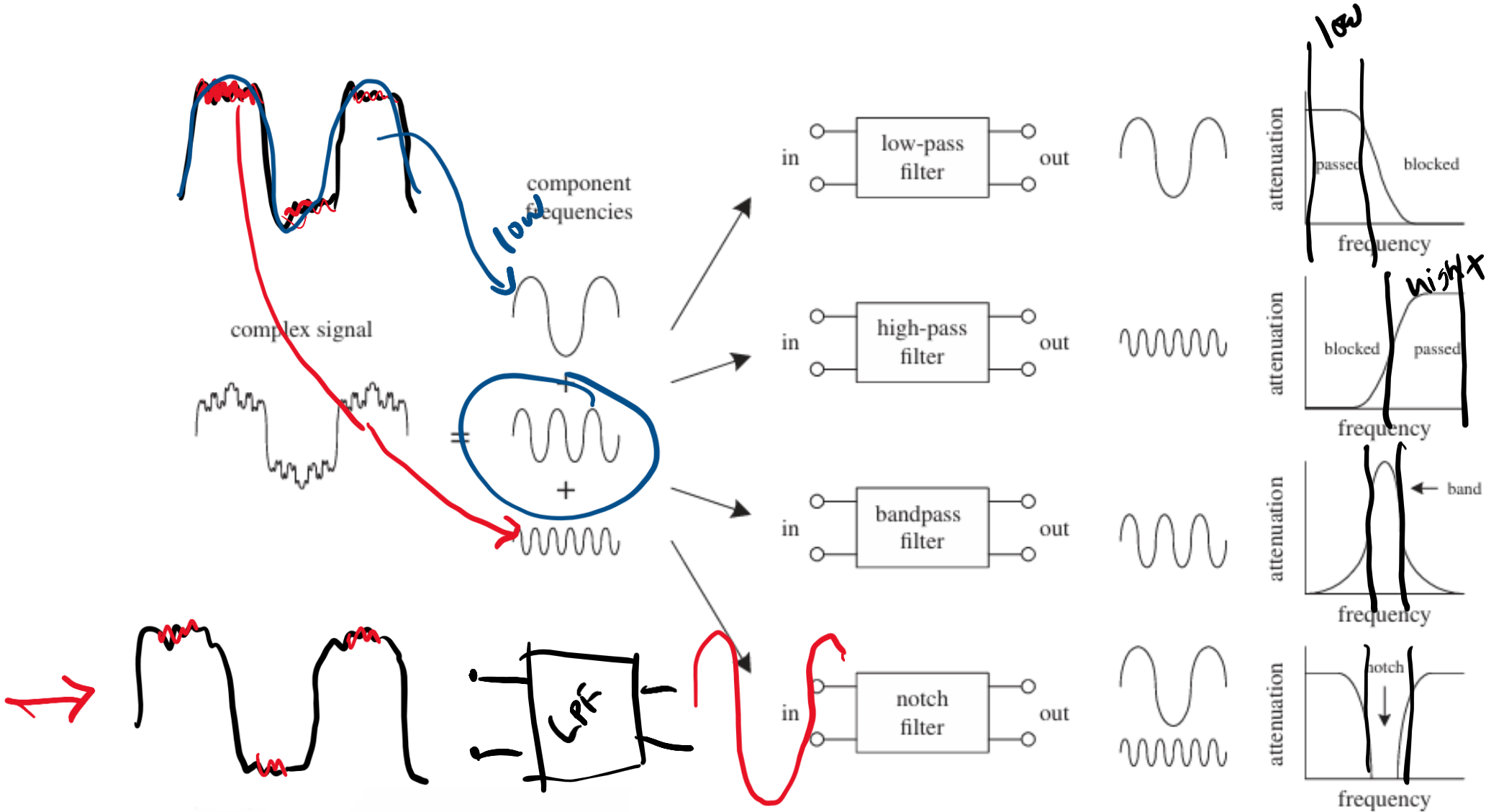


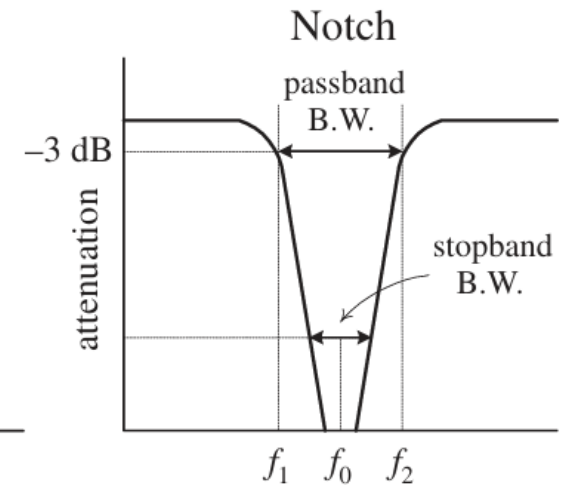
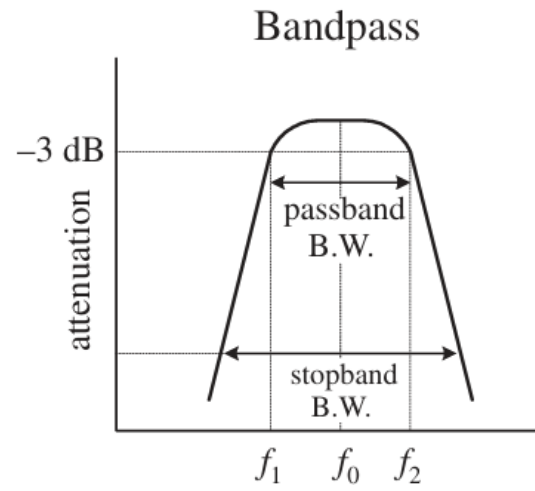
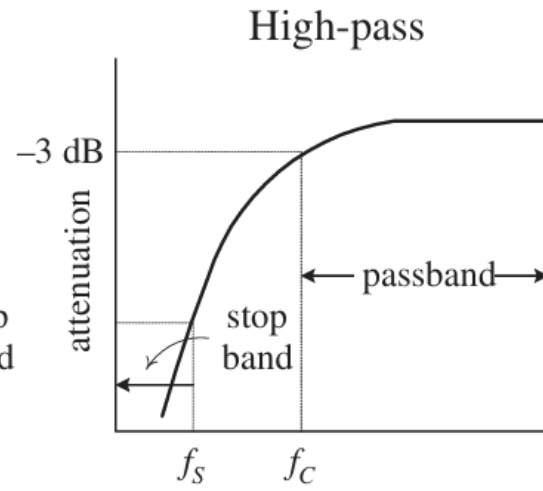
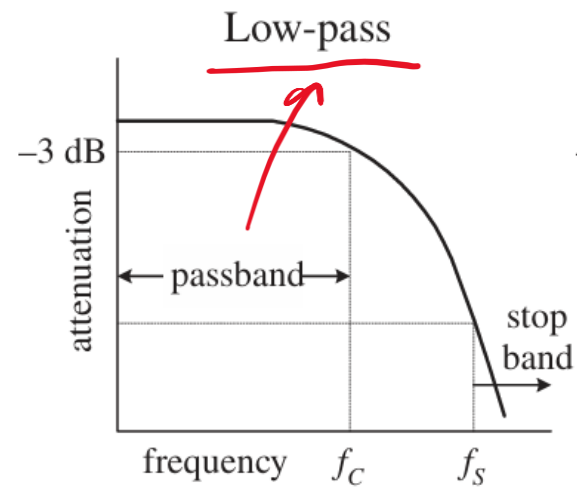
What makes a square wave



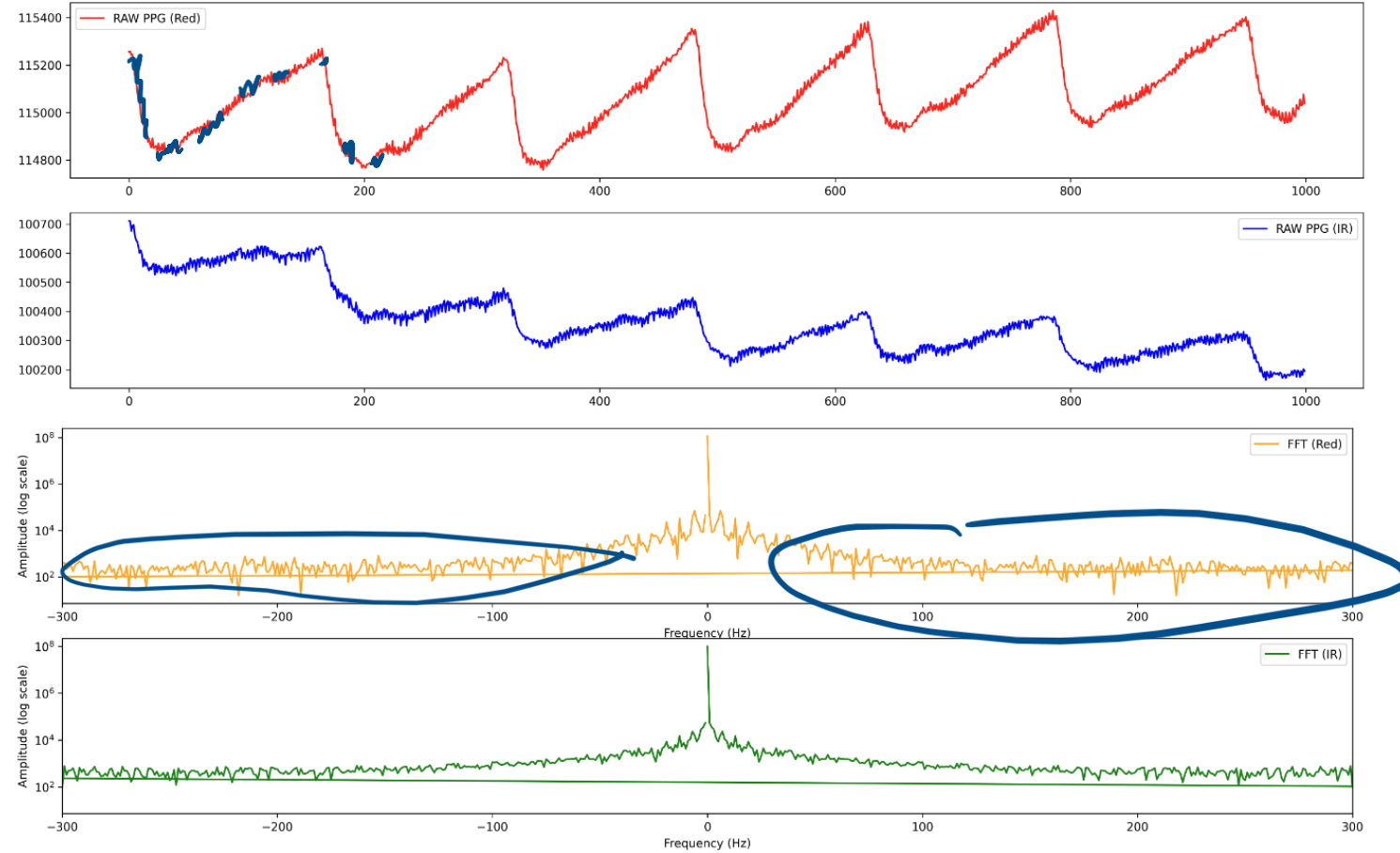
$f \rightarrow$

Filters

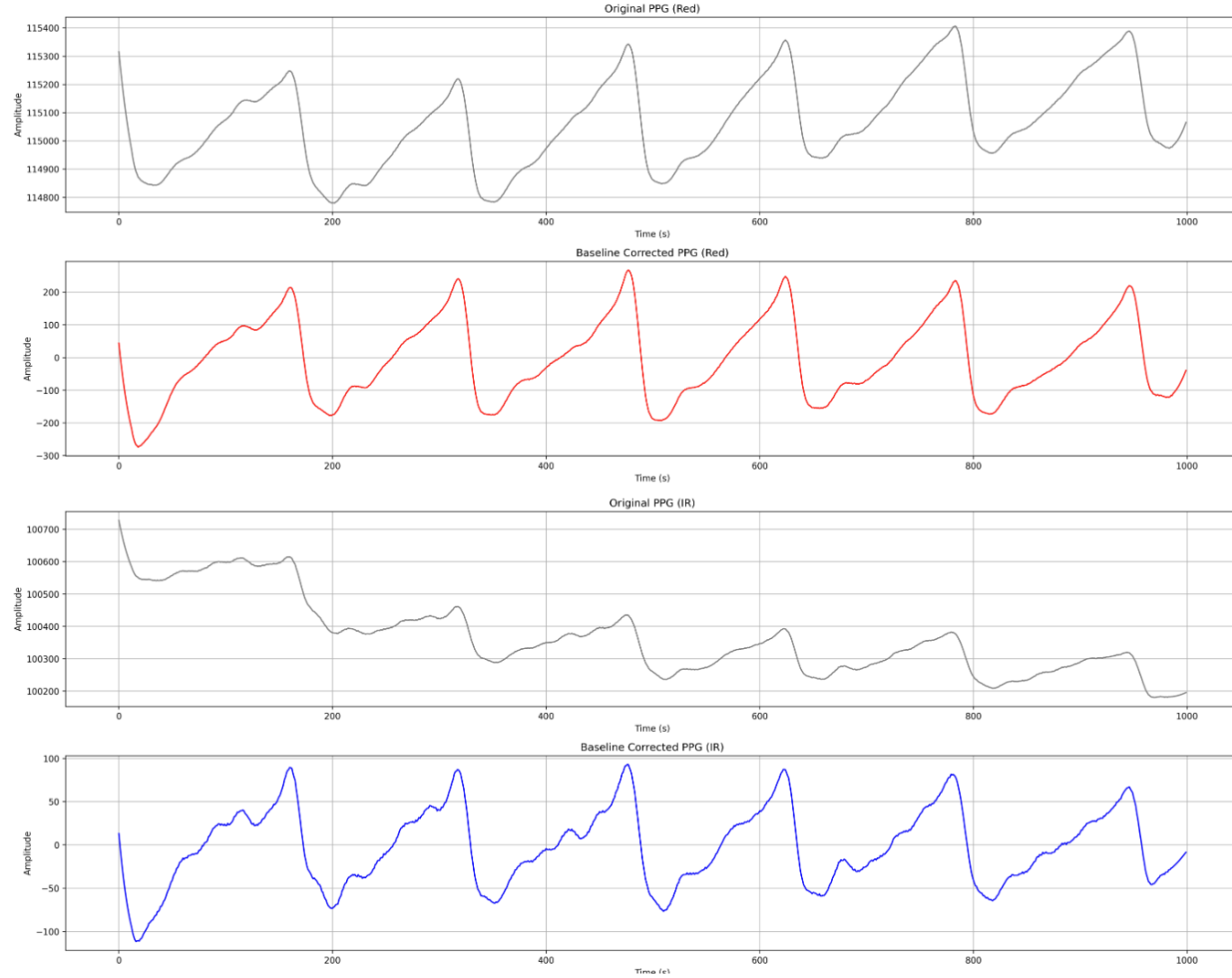




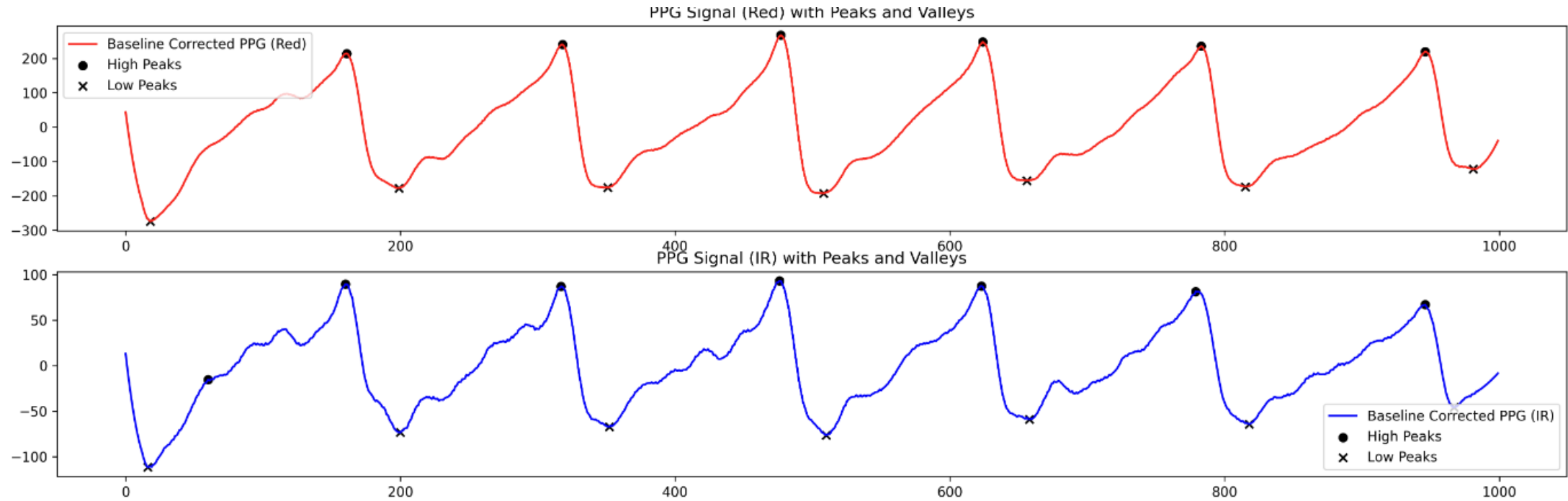
Going back to the PPG data collected



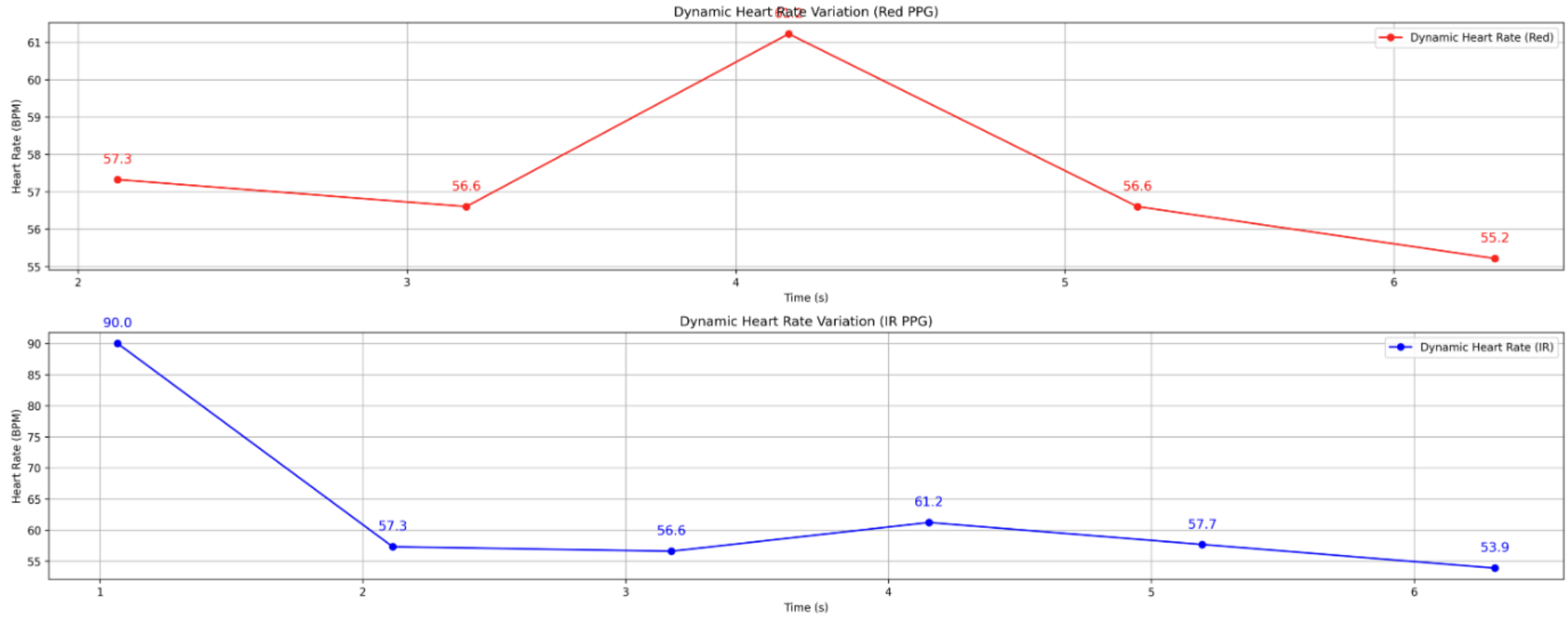
Drift correction



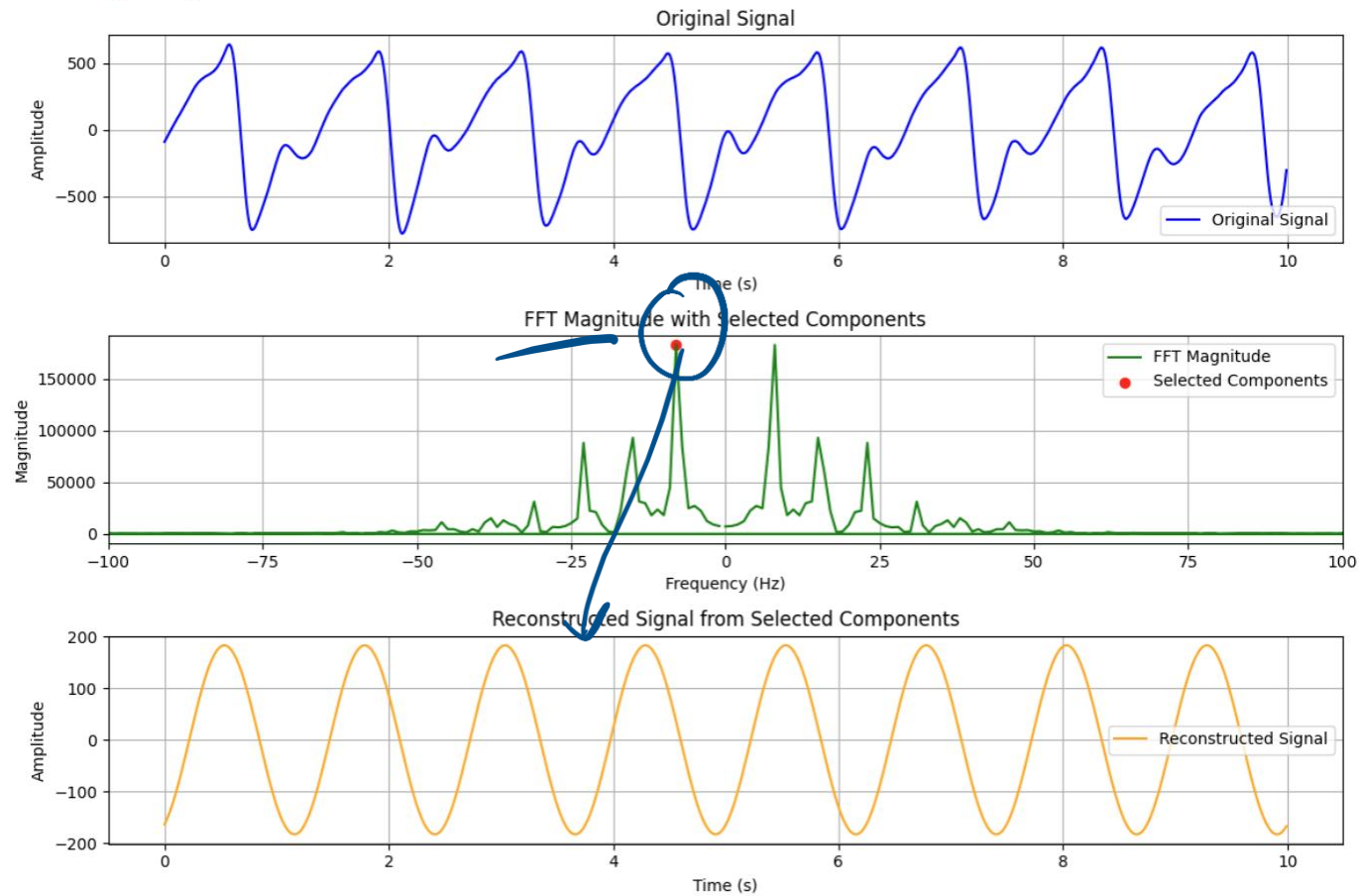
Peak detection



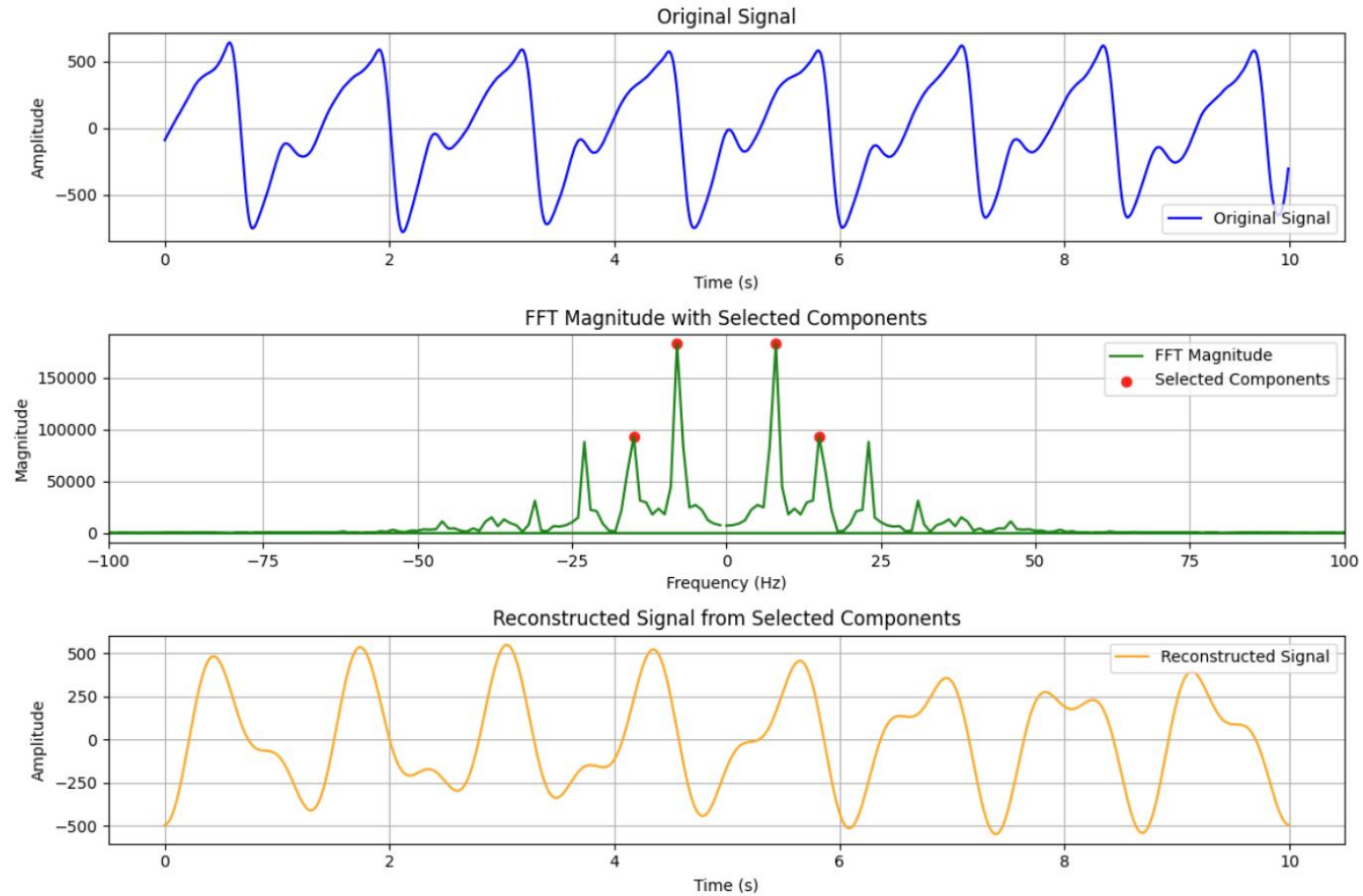
Heart rate



Using single component to find heart rate



4 components in the PPG signal – signal loss



Taking all the components provides the original signal

